

Dear Editors,

I am writing to submit the manuscript titled “Computational evidence for a distributed AT-based regulatory mechanism enabling de novo gene birth in *Pandoravirus salinus*” for consideration as a Research Article in *Genome Biology and Evolution*.

Pandoravirus salinus possesses the largest known viral genome (2.47 Mb), with over 92% of its genes having no detectable homologs outside the Pandoraviridae. While comparative genomic studies by the Claverie group have established that these genes likely arise de novo from non-coding sequence, the genomic infrastructure enabling this process has remained unclear. Using a 12-module computational pipeline applied to a single genome, we first establish that *P. salinus* is compositionally the most uniform genome in a four-virus comparison set despite being the largest (GC coefficient of variation 6.1%), with ORFan genes indistinguishable from annotated genes across all codon usage metrics arguing against recent patchwork horizontal gene transfer. We then identify a distributed AT-based regulatory system that pre-loads intergenic regions with promoter and terminator signals, resolving how proto-genes can be transcribed regardless of their genomic position. We establish a quantitative five-tier proto-gene continuum supported by gene family analysis (96.8% singletons) and protein structure prediction (ESM-2 embeddings), and demonstrate through genome-wide BLASTp analysis that annotated genes are 7.2-fold more likely than hypothetical genes to retain external homologs (OR = 7.19, 95% CI: 3.5–13.6, $p = 7.79 \times 10^{-12}$) providing direct evidence for an age-dependent homology gradient consistent with the de novo gene birth model.

I believe this manuscript is well suited for GBE because it addresses a fundamental question in genome evolution how new genes arise from non-coding DNA through computational methodology that complements existing comparative approaches. The single-genome analytical framework we present provides an independent line of support for the de novo gene birth hypothesis and identifies the mechanistic substrate that enables it. This work should be of interest to GBE’s readership in evolutionary genomics, giant virus biology, and de novo gene emergence.

This manuscript has not been published and is not under consideration elsewhere. All analysis scripts, data, and a statistics audit file are publicly available at <https://github.com/mathamagician/pandoravirus-denovo-genebirth> and archived at <https://doi.org/10.5281/zenodo.19046142>.

Disclosure of AI tool use: In accordance with GBE policy, I wish to disclose that AI tools (Anthropic’s Claude) were used to assist with code development, data-processing workflow design, analytical planning, and editorial refinement of manuscript text. This use is also disclosed in the Materials and Methods section of the manuscript. All analyses, statistical methods, interpretations, and final manuscript content were reviewed and validated by the author. The author takes full responsibility for the accuracy and integrity of the work.

Thank you for your consideration.

Sincerely, Edward Bradford